

Computational Design and Screening of De Novo Antimicrobial Peptides Using Generative AI

P Sai Lekhya

Computer Science Of Engineering
R. V. College Of Engineering,
Bengaluru, Karnataka-560059, India
psailekhya.cs24@rvce.edu.in

Parinitha M

Computer Science Of Engineering
R. V. College Of Engineering
Bengaluru, Karnataka-560059, India
parinitham.cs24@rvce.edu.in

Dr. Ajeet Kumar Srivastava

Biotechnology (Assistant Professor)
R. V. College Of Engineering
Bengaluru, Karnataka-560059, India
ajeeth@rvce.edu.in

Abstract—

*The escalating crisis of antimicrobial resistance necessitates the rapid development of novel therapeutic and diagnostic modalities to address global health challenges. This study presents a dual-pronged computational framework that integrates generative artificial intelligence for drug discovery with synthetic biology for point-of-care diagnostics. First, a de novo design pipeline for antimicrobial peptides is established using protein language models to generate novel peptide sequences independent of natural templates. These candidates undergo rigorous computational screening for antimicrobial potential and toxicity profiling, followed by structural validation to ensure membrane-targeting efficacy against multidrug-resistant pathogens. Second, a CRISPR-Cas12a-based lateral flow assay is proposed for the detection of *Leptospira interrogans*, the causative agent of leptospirosis. By leveraging the collateral cleavage activity of the Cas12a enzyme, this design enables highly specific and instrument-free diagnostics suitable for resource-limited settings. Guide RNA specificity is validated against human and microbiome genomes to prevent off-target effects and ensure diagnostic accuracy. This integrated approach demonstrates the feasibility of using advanced computational tools to accelerate the development of next-generation anti-infective strategies. The proposed workflow offers a scalable blueprint for combating resistant bacteria and improving diagnostic access for neglected tropical diseases, highlighting the transformative potential of interdisciplinary bioengineering in modern medicine.*

Keywords - Antimicrobial peptides, antimicrobial resistance, generative AI, protein language models, de novo drug discovery, CRISPR-Cas12a, lateral flow assay, leptospirosis, point-of-care diagnostics, synthetic biology.

I. INTRODUCTION

Antimicrobial resistance (AMR) represents one of the most pressing public health threats of the 21st century, with projections indicating that drug-resistant infections could cause ten million deaths annually by 2050, surpassing cancer as a leading cause of mortality [1][6]. The stagnation in the development of conventional small-molecule antibiotics, combined with the rapid evolution of multidrug-resistant pathogens, has created an urgent demand for innovative therapeutic and diagnostic solutions [7][8]. Traditional antibiotic discovery pipelines are often labor-intensive, expensive, and limited by the finite diversity of natural product libraries, prompting a paradigm shift toward data-centric, artificial intelligence (AI)-driven approaches [9]. Among the most promising alternatives are antimicrobial peptides (AMPs), short amino acid sequences that exhibit broad-spectrum activity through membrane-disrupting mechanisms that are less prone to inducing bacterial resistance

[10][11]. However, the identification of potent AMPs from the vast combinatorial sequence space remains a significant challenge, traditionally relying on trial-and-error screening or modest modifications of existing natural peptides [12][13].

Recent advances in deep learning and protein language models have revolutionized the *de novo* design of functional proteins, enabling the generation of novel AMP sequences that do not exist in nature [12]. Models such as ProtGPT2 and ESM-2 leverage transformer architectures trained on extensive protein databases to capture complex sequence-structure-function relationships, allowing for the creation of diverse and druggable peptide candidates [4][14]. These generative frameworks can be integrated with robust computational screening pipelines, including AMP likelihood predictors like AMPscanner and toxicity assessment tools such as ToxinPred, to efficiently filter candidates for high efficacy and low human cell toxicity [3][15]. Furthermore, structural prediction tools like ESMFold provide critical insights into the three-dimensional conformations of these synthetic peptides, facilitating the rational design of membrane-targeting agents against Gram-negative bacteria [16][4]. This AI-driven workflow not only accelerates the discovery process but also expands the chemical repertoire available for combating multidrug-resistant organisms such as the ESKAPE pathogens [17][18].

Parallel to the need for new therapeutics is the critical requirement for rapid, accessible diagnostics, particularly for neglected tropical diseases that disproportionately affect low-resource regions. Leptospirosis, a zoonotic disease caused by *Leptospira* species, is endemic in tropical and flood-prone areas yet remains significantly underdiagnosed due to the lack of affordable, point-of-care testing options [19]. Current diagnostic methods often require specialized laboratory infrastructure, delaying treatment and contributing to higher morbidity and mortality rates. CRISPR-Cas systems, originally identified as bacterial adaptive immune mechanisms, have been repurposed into powerful molecular diagnostic platforms known as SHERLOCK and DETECTR [5]. Specifically, the Cas12a enzyme exhibits a unique "collateral cleavage" activity, where target recognition triggers indiscriminate cutting of nearby single-stranded DNA reporters, resulting in massive signal amplification without the need for complex instrumentation [5][19].

By combining isothermal amplification with CRISPR-Cas12a detection, it is possible to develop lateral flow assays that offer high sensitivity and specificity comparable to polymerase chain reaction (PCR) but with the simplicity of a pregnancy test strip [5]. The design of such assays requires careful selection of conserved pathogen-specific target sequences and the engineering of guide RNAs that minimize off-target binding to human or commensal bacterial DNA [5]. This study integrates these two frontier technologies—generative AI for peptide design and CRISPR-based diagnostics—to propose a comprehensive strategy for addressing the AMR crisis. We present a computationally validated pipeline for *de novo* AMP generation and a detailed architectural design for a CRISPR-Cas12a lateral flow assay for leptospirosis, demonstrating how interdisciplinary bioengineering can yield impactful solutions for global health challenges [9][20].

II. METHODOLOGY

A. Overview of the Pipeline

This study presents a modular computational pipeline for the *de novo* design and screening of antimicrobial peptides (AMPs) using generative AI. The pipeline comprises five sequential stages: peptide sequence generation, physicochemical filtering, antimicrobial activity prediction, toxicity screening, and three-dimensional structure prediction and visualization. All tools used are open-source or publicly accessible, requiring no proprietary software or wet-lab infrastructure.

B. Peptide Sequence Generation

Novel peptide sequences were generated using ProtGPT2, a transformer-based protein language model pre-trained on the UniRef50 database. The model was executed via the Hugging Face Transformers library on Google Colab. A total of 50 candidate sequences were generated using top-k sampling with a temperature parameter set to 1.0 to promote sequence diversity. Generated sequences were decoded and stored in FASTA format for downstream processing.

C. Physicochemical Filtering

Raw outputs from ProtGPT2 were subjected to quality filtering to retain biologically relevant candidates. Sequences were filtered based on the following criteria:

- Length between 10 and 60 amino acids

- Absence of ambiguous or non-standard amino acid characters (e.g., X, B, Z)

- Minimum net positive charge of +1, consistent with known membrane-active AMPs

Sequences failing any criterion were discarded from further analysis.

D. Antimicrobial Activity Prediction

Filtered sequences were submitted to AMP Scanner v2, which provides two complementary classification models: a Random Forest and MARS-based AMP classifier, and a Deep Neural Network (DNN)-based AMP classifier. Both models were applied independently to each candidate peptide, with each assigning a binary prediction of AMP or Non-AMP along with a corresponding probability score.

A consensus-based classification scheme was adopted to improve prediction reliability. Peptides predicted as AMP by both models were designated as Gold candidates, indicating high-confidence antimicrobial activity. Peptides predicted as AMP by only one of the two models were designated as Silver candidates, reflecting moderate confidence. Peptides classified as Non-AMP by both models were excluded from further analysis.

Additionally, the Antimicrobial Peptide Database (APD3) was used as a complementary reference to cross-validate physicochemical properties such as charge, hydrophobicity, and hydrophobic moment against experimentally validated AMP benchmarks.

E. Toxicity Screening

Peptides predicted as AMPs were further evaluated for mammalian cytotoxicity using ToxinPred, a support vector machine (SVM)-based tool developed by the CSIR-Institute of Microbial Technology. ToxinPred computes an SVM score where negative values indicate non-toxic predictions. Only peptides classified as Non-Toxic were retained as viable candidates to ensure therapeutic safety.

hydrophobicity, and hydrophobic moment against experimentally validated AMP benchmarks.

F. Three-Dimensional Structure Prediction

The top AMP candidates that passed both antimicrobial activity prediction and toxicity screening were subjected to three-dimensional structure prediction using ESMFold v1, implemented as a Google Colab notebook with GPU acceleration (Tesla T4). ESMFold is an end-to-end single-sequence structure prediction model developed by Meta AI, built upon the ESM-2 protein language model, and does not require multiple sequence alignments (MSAs).

The ESMFold environment was set up by installing required dependencies including OpenFold, omegacnf, pytorch_lightning, biopython, einops, and py3Dmol. Model parameters were downloaded from the ColabFold Steinegger Lab repository prior to inference. Each candidate peptide sequence was provided as input, and the model generated atomic coordinate predictions in PDB format.

For each candidate, the following confidence metrics were extracted and analyzed:

- **pLDDT score** — per-residue local distance difference test score (0–100), plotted across sequence positions; scores above 70 indicate high structural confidence
- **Predicted Aligned Error (PAE)** — inter-residue positional uncertainty visualized as a 2D matrix; lower values (blue regions) indicate high inter-domain confidence

- **Contact map** — short-range residue-residue contacts predicted from the structure module's distogram logits, with contact threshold set at 8 Å

Predicted PDB structures and confidence plots were exported as a compressed archive and downloaded for downstream visualization. Candidates with a mean pLDDT > 60 across the full sequence were retained for structural characterization.

G. Structure Visualization and Analysis}

Predicted PDB structures were visualized using PyMOL (Schrödinger LLC). Surface electrostatic maps were generated to assess amphipathicity, a key structural property of membrane-disrupting AMPs. The blue (positively charged) and white (hydrophobic) surface regions were examined to identify the characteristic cationic-amphipathic pattern associated with bacterial membrane disruption.

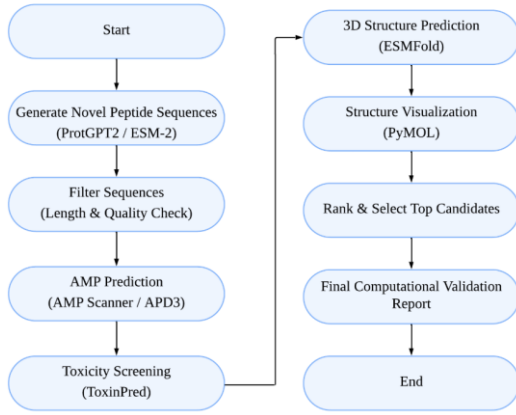


Fig 1: Methodology Flowchart

III. RESULTS AND DISCUSSIONS

A. Peptide Generation

A total of 50 novel peptide sequences were generated using ProtGPT2. The generated sequences ranged in length from 18 to 51 amino acids, consistent with the typical size range of known antimicrobial peptides. All sequences were subjected to physicochemical filtering, retaining those within a length of 10–60 amino acids with a minimum net positive charge of +1.

B. Antimicrobial Activity Prediction

The filtered sequences were evaluated using two complementary models in AMP Scanner v2: a Random Forest and Multivariate Adaptive Regression Splines (MARS)-based classifier, and a Deep Neural Network (DNN)-based classifier. A consensus-based scheme was applied, wherein peptides predicted as AMP by both models were designated Gold candidates, and those predicted by only one model were designated Silver candidates.

Of the 50 generated sequences, 24 were classified as AMP by the DNN-based classifier (Version 1), with AMP probability scores ranging from 0.50 to 1.0. From the RF/MARS-based classifier (Version 2), only Candidate 44 was predicted as AMP by both the RF and MARS sub-models, while Candidates 21, 42, and 50 were predicted as AMP by the

MARS sub-model alone. Applying the consensus rule, Candidate 44 was the only Gold candidate confirmed by both classifiers. Candidates 21 and 42 were classified as Silver, having achieved high-confidence DNN scores of 0.9999 and 0.9993 respectively, combined with MARS-based AMP prediction. Candidate 44 achieved a perfect DNN score of 1.0 and was predicted AMP by both Version 1 and Version 2 models, making it the strongest candidate overall.

Sequence ID(s)	Predicted Class	Prediction Probability
AMP_candidate_1 MDTTVALILLITFIIFVIVFLKLNK...	AMP	0.9633
AMP_candidate_2 MDNQNDNTHLDIAKKIEESTQDKQENK	Non-AMP	0.0008
AMP_candidate_3 MADKNTYIDYNYVQDLLDQFERQNYISKK...	Non-AMP	0.0201
AMP_candidate_4 MSNIIISKDQLDLCNRLIKESEQNQYSLN...	Non-AMP	0.0005
AMP_candidate_5 MPRTNRKRKKRKELKSKKSARKEEKRE...	Non-AMP	0.0169
AMP_candidate_6 MNDDIHRSCSDIYVSFYQLHNFYFFIK...	AMP	0.5309
AMP_candidate_7 MLLGKKIWESREFYFFYNTMSNDKILKL...	Non-AMP	0.0002
AMP_candidate_8 MGTHIRHTFDPGTCLCGGRDACAHPHTGW...	AMP	0.9389
AMP_candidate_9 MLNFDFIALISAVLVGFVVEVYQYIKK...	Non-AMP	0.1989
AMP_candidate_10 MTWLFVLLACVPVAIFLVALAVRWSDRYD...	AMP	0.6334
AMP_candidate_11 MTCSIGVARFPADGADADTLVKAADSALYR...	Non-AMP	0.1446
AMP_candidate_12 MALSDAQIALISQYKSAQVSEQTIAKAFGL...	Non-AMP	0.003
AMP_candidate_13 MIHLIIIALVVALVYLYKSKNGKSCSSC...	AMP	0.9991

Fig2: AMPscanner Classification Results for Candidate Peptides

C. Toxicity Screening

All three top candidates — Candidate 21, Candidate 42, and Candidate 44 — were evaluated using ToxinPred. The SVM-based toxicity scores were −1.28, −0.98, and −0.23 respectively, all negative values indicating Non-Toxic classification. Candidate 21 exhibited the most strongly negative SVM score, suggesting the lowest predicted mammalian cytotoxicity among the three. All candidates were confirmed as Non-Toxic and retained for structural analysis.

The ToxinPred results also provided physicochemical properties for the candidates. Candidate 21 (MSTKAKKKLIKKKKLLKQKNLKPCLKKKPLNLQK KIKTKNTKKKKKK) exhibited an exceptionally high net charge of +26 and a hydrophilicity value of 1.25, consistent with a strongly cationic, membrane-disruptive AMP profile. Candidate 42 (MGWLYLVLGVVIVVGIIIFVVAKTSGKPKTEKGNK) showed a moderate charge of +4 and high hydrophobicity of 0.86, reflecting a predominantly hydrophobic helical character. Candidate 44 (MPTGQQFSIRYPRNNGPDRGCGWGRFCRCL) carried a neutral charge of 0 and a hydrophobicity of −0.91, with a cysteine-rich sequence suggestive of a defensin-like structural scaffold.

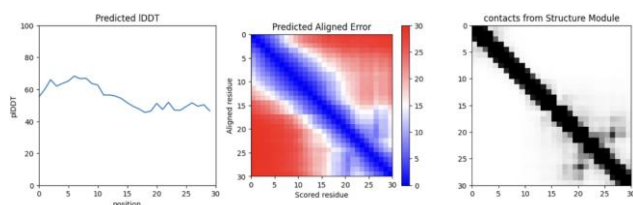


Fig9: Predicted IDDT, Predicted Aligned Error, and Residue Contact Map

E. Surface Electrostatic Analysis

PyMOL surface visualization revealed distinct amphipathic profiles for all three candidates.

Candidate 21 displayed an extensively blue (cationic) surface across its helical length, consistent with its charge of +26, suggesting strong electrostatic attraction to the negatively charged bacterial membrane.

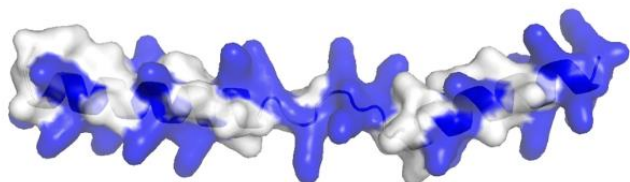


Fig 10: PyMOL Surface Renderings of Candidate 21(MSTKAKKKLIKKKKLLKQKNLKPCLKKKPLNLQKKIKTKNTKKKKKK)

Candidate 42 showed a predominantly white (hydrophobic) surface with localized blue patches, indicative of an amphipathic helix capable of inserting into lipid bilayers.

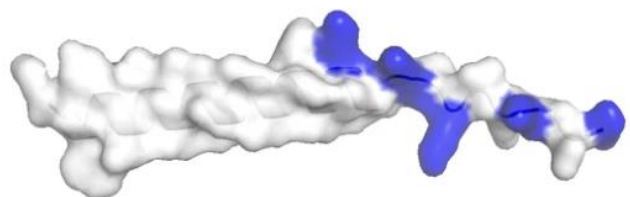


Fig 11: PyMOL Surface Renderings of Candidate 42 (MMGWLYLVLGVVIVVGILFVVAKTSGKPKTEKGNK)

Candidate 44 exhibited a mixed surface with cationic patches distributed around a largely neutral core, consistent with defensin-type membrane interaction mechanisms.

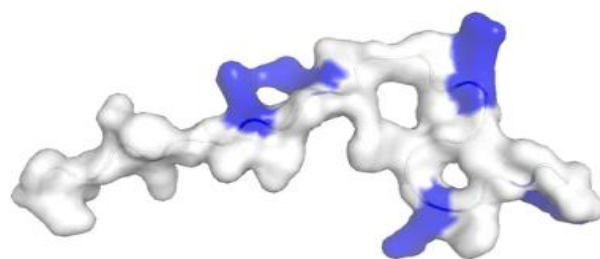


Fig 12 : PyMOL Surface Renderings of Candidate 44 (MPTGQQFSIRYPRNNGPDRGCGWGRFCRCL)

F. Discussion

The three-stage screening pipeline — AMP prediction, toxicity filtering, and structural analysis — successfully narrowed the initial 50 generated candidates to three high-priority leads. The convergence of high DNN-based AMP probability scores, negative ToxinPred SVM scores, and favorable structural predictions across Candidates 21, 42, and 44 provides strong computational evidence for their potential antimicrobial activity.

Candidate 21 is particularly notable for its extreme cationic character, a property that strongly correlates with the ability to electrostatically target anionic bacterial membranes. Candidate 42 resembles transmembrane-active hydrophobic helical AMPs and may operate via a carpet or toroidal pore mechanism. Candidate 44, despite its lower pLDDT confidence, exhibits a cysteine-rich motif that is characteristic of defensin-class peptides, which are among the most potent naturally occurring AMPs.

A key limitation of this study is the absence of wet-lab validation. All predictions are computational and dependent on the accuracy of the underlying models. AMP Scanner v2 and ToxinPred are trained on known sequences and may not fully capture the novelty space explored by ProtGPT2-generated peptides. ESMFold, while highly capable for monomeric single-sequence prediction, cannot model disulfide bonding or membrane-induced conformational changes. Future work should include molecular dynamics simulations, docking studies against bacterial membrane models, and experimental MIC (Minimum Inhibitory Concentration) assays to validate these candidates in vitro.

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